Nicholas Di

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Research Interests and Skills

Numerical Optimization ● Statistical Learning

Languages: Python; R; MATLAB

ML / Stats: PyTorch; TensorFlow; Scikit-learn; NumPy; Pandas; CVXPy; Causal Inference; Uncertainty Quantification

Optimization & Computation: Proximal Algorithms; Stochastic Optimization; Convex & Nonconvex Optimization; Monte Carlo Methods; Scientific Computing

Scientific Computing & Tools: Linux; Git; SLURM; HPC; Docker; Tetrad

Education

Rice University — **Houston, TX, USA**
PhD Statistics (GPA: 3.96) — *Aug 2023 – May 2028 (Expected)*

Macalester College — **Saint Paul, MN, USA**
BA Economics (Honors); BA Applied Mathematics / Statistics (GPA: 3.96) — *Aug 2018 – Dec 2022*

Publications

- **Di N.**, Chi E., Wu Fung S. (2026). Operator Splitting with Hamilton-Jacobi-based Proximals. *International Conference of Machine Learning*.
- **Di N.**, Chi E., Wu Fung S. (2025). A Monte Carlo Approach for Nonsmooth Convex Optimization via Proximal Splitting Algorithms. *Short version accepted by NeurIPS Optimization for Machine Learning Workshop*.
- Artman C., **Di N.**, Perkins S. (2025). IFlowNets: Extending Generative Samplers to Learn Strategies in Incomplete Information Games. *Short version accepted by NeurIPS Dynamics at the Frontiers of Optimization, Sampling, and Games Workshop*.
- Shaw C., Williams C., Tan T., Illera D., **Di N.**, Shulman J., Belmont J. (2025). The Causal Pivot: A Structural Approach to Genetic Heterogeneity and Variant Discovery in Complex Diseases. *The American Journal of Human Genetics*.
- **Di N.** (2023). Gendered Labor Market Outcomes During COVID-19: Evidence from Early Withdrawal of Federal Pandemic Unemployment Compensation. *The Developing Economist*. **(Best Undergraduate Paper Award)** [Paper] [News]

Research Experience

Triangulated Fusion Models for Spatial Regime Detection — **Jan 2026 – Present**
Advised by Dr. Huixia Judy Wang, Rice University

- Developed the Triangulated Fusion Model, a spatial statistical framework combining bivariate penalized splines over triangulations with regime detection to analyze heterogeneous responses to decarbonization drivers.
- Engineered a computationally scalable solver that maps local Bernstein coefficients into a global monomial coordinate system, achieving a 125x speedup and lower mean integrated squared error compared to standard spatial clustering models.

Stochastic Methods for Non-Smooth Optimization — **Sep 2024 – Present**
Advised by Dr. Eric Chi, Rice University

- Developed Monte Carlo–based splitting algorithms for non-smooth convex optimization problems with intractable proximal operators, enabling scalable computation for large, complex decision systems.
- Established convergence guarantees for stochastic proximal splitting methods under multiple noise models, strengthening the reliability of optimization under uncertainty in computationally constrained settings.
- Positioned stochastic optimization methods for downstream use in large-scale scientific and engineering applications where exact solvers are computationally impractical.

AI Applications in Game Theory — **May 2025 – Aug 2025**
Advised by Dr. Conor Artman & Dr. Scott Perkins, Lawrence Livermore National Lab

- Extended Generative Flow Networks (GFlowNets) to incomplete-information games, developing learning algorithms for sequential decision-making under uncertainty.
- Designed and executed large-scale computational experiments on SLURM-managed high-performance computing clusters, improving throughput and reproducibility for advanced ML research.
- Developed theoretical connections between generative sampling methods and counterfactual regret minimization, linking machine learning, optimization, and strategic decision systems.

PRS–RV Causal Pivot: A Segmentation Approach to Complex Disease Genetics — May 2024 – Aug 2024

Advised by Dr. Chad Shaw, Baylor College of Medicine

- Developed a structural equation model categorizing diseases into four types using Polygenic Risk Scores (PRS) and rare variants (RV), showing that RV–PRS exclusion can aid in disease discovery.
- Tested the model on UK Biobank data for hypercholesterolemia, breast cancer, and Alzheimer's disease, finding a significant negative correlation between RV and PRS in affected individuals.

Gendered Labor Market Outcomes During COVID-19 — Jun 2022 – Aug 2022

Advised by Dr. Amy Damon, Macalester College

- Conducted independent econometric research on the gender-specific effects of ending Federal Pandemic Unemployment Compensation, analyzing how policy shocks affected labor-market outcomes.
- Collected and analyzed data from the U.S. Census Bureau; collaborated with a Federal Reserve researcher to validate findings and translate statistical evidence into policy-relevant conclusions.

Causal Inference — May 2019 – Aug 2019

Advised by Dr. Leslie Myint, Macalester College

- Developed a causal diagram using R and Tetrad search algorithms, applying causal inference techniques to enhance the reliability of biological research findings.
- Spearheaded a meta-research initiative analyzing methodological rigor in cancer biology, developing text-mining pipelines to systematically evaluate 80+ research abstracts from PLoS and Semantic MEDLINE databases.

Applied Research & Industry Experience

Data Science Intern — May 2025 – Aug 2025

Lawrence Livermore National Laboratory, Livermore, CA, USA

- Engineered novel Generative Flow Network architectures for strategic decision-making in imperfect-information environments, combining generative AI with optimization for complex autonomous systems.
- Built and scaled experiment pipelines for large-batch model training and hyperparameter tuning on high-performance computing infrastructure.
- Led an interdisciplinary student team in delivering an applied computer-vision solution for robotics workflows, demonstrating experience at the intersection of AI, automation, and real-world systems.

Research Assistant Intern — Jun 2022 – Aug 2022

The Brattle Group, Boston, MA, USA

- Developed econometric and optimization models for electricity demand forecasting and grid-related decision problems, contributing quantitative insight relevant to energy-system planning and operations.
- Applied mathematical optimization in MATLAB to large-scale energy market and infrastructure questions, linking rigorous analytics with practical operational decisions.
- Collaborated with academic experts on causal-inference and policy-oriented modeling for high-impact applied problems.

Analyst — Jan 2023 – Jun 2023

Ernst & Young, Quantitative Economics and Statistics, Washington, D.C., USA

- Performed econometric analysis, predictive forecasting, and fiscal-impact modeling for business and government clients, translating quantitative evidence into policy- and strategy-relevant recommendations.
- Supported analyses related to ESG, public policy, taxation, relocation, and vaccine distribution, building experience in how quantitative methods inform large-scale societal and regulatory decisions.

Analyst Intern — Jun 2021 – Aug 2021

Ernst & Young, Quantitative Economics and Statistics, Washington, D.C., USA

- Maintained databases for Medicare case management, implementing robust data-governance frameworks and compliance tracking systems to ensure adherence to healthcare-accredited standards.
- Supported senior researchers in economic-policy analysis projects, gaining practical experience in regulatory impact assessment, survey methodology design, and statistical sampling while contributing to client deliverables.

Business Development Manager — Aug 2020 – May 2021

NutriKarma, New York, NY, USA

- Expanded the NutriKarma team by 11 employees and contributed to project management across app wire-flows, data sets, and marketing.
- Led meetings with potential angel investors, securing interest and building strategic partnerships with business leaders, dietitians, and personal trainers to advance market presence.

Business Analyst Intern **Jun 2020 – Aug 2020**

U.S. Bank, Minneapolis, MN, USA

- Facilitated projects on classification of product types for document custody services and performed in-depth data and business analysis for automation of the loan certification process.

Business Data Analyst **Oct 2018 – May 2019**

Minnesota Association of Volunteer Administration, Minneapolis, MN, USA

- Wrote R scripts to compute regression models on membership recruitment data for 500+ clients.
- Developed outreach plans for local organizations and sponsors, and updated internal resources to keep information and schedules current.

Presentations

- **Operator Splitting with Hamilton-Jacobi-based Proximals**, Invited Talk, SIAM Mathematics of Data Science — 2026
- **Operator Splitting with Hamilton-Jacobi-based Proximals**, Contributed Talk, Joint Statistical Meetings — 2026
- **IFlowNets: Extending Generative Samplers to Learn Strategies in Incomplete Information Games**, SIAM TX-LA — 2025
- **The Polygenic Score Rare Variant Causal Pivot**, American Society of Human Genetics — 2024
- **Spatial Analysis of Lead Levels in the Twin Cities Metro Area**, Capstone Seminar, Macalester College — 2022
- **Economics Independent Research**, Midwest Economics Association — 2022
- **Causal Inference Research**, Jr. Faculty-Hub Summer Research Fund Meeting, Macalester College — 2019

Awards

- **Rice Statistics Department Travel Award** — 2024, 2025
- **AAAS Annual Conference Statistics Travel Award** — 2024
- **Dean's Prize**, Rice University — 2023–2025
- **3M Scholar Award**, Macalester College Economics Department — 2022
- **Best Undergraduate Paper Award**, Midwest Economics Association Conference — 2022
- **Summa Cum Laude**, Macalester College — 2022
- **Phi Beta Kappa**, Macalester College — 2022
- **Dean's List, All Semesters**, Macalester College — 2018–2022

Professional Service

- **Reviewer**, International Conference on Machine Learning (ICML) — 2026
- **Invited Reviewer**, Conference on Neural Information Processing Systems (NeurIPS) — 2026

Teaching Activities

Rice University

- Teaching Assistant — STAT 613: Statistical Machine Learning (Graduate)
- Teaching Assistant — STAT 413: Statistical Machine Learning (Undergraduate)
- Teaching Assistant — STAT 425: Introduction to Bayesian Inference (Undergraduate)
- Teaching Assistant — STAT 405/605: Data Science in R (Undergraduate/Graduate)

Macalester College

- Teaching Assistant — COMP 123: Introduction to Computer Science
- Teaching Assistant — STAT 155: Introduction to Statistical Modeling
- Teaching Assistant — ECON 194: Calculus-Based Principles of Economics
- Teaching Assistant — STAT 253: Statistical Machine Learning
- Supplemental Instructor — ECON 381: Econometrics

University Service

- **Statistics Student Seminar Organizer**, Rice University — 2024
- **STAT 450 Project Judge**, Rice University — 2022
- **Graduate Student Representative**, Rice University — 2023